The Power of Vaccination

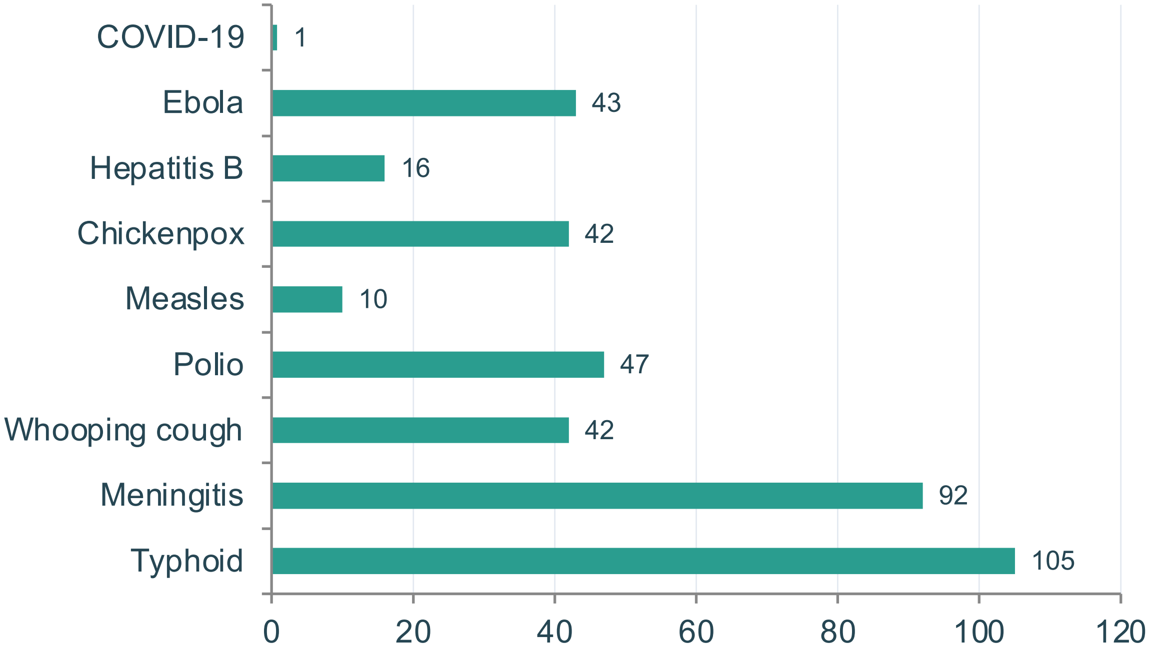
From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of vaccine impact across two centuries (1796–2020)

Source: Our World in Data

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure



COVID-19: <1 year

Fastest vaccine development in history, enabled by mRNA technology and genomic tools

Typhoid fever took 105 years (1884→1989)

Source: Our World in Data

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases (per million/yr)	Post-Vaccine Cases (per million/yr)	Case Reduction	Pre-Vaccine Deaths (per million/yr)	Death Reduction
Diphtheria	158	0	**100%**	13.7	**100%**
Smallpox	250	0	**100%**	2.9	**100%**
Acute polio	141	0	**100%**	10	**100%**
Measles	3,044	0.2	99.99%	2.5	**100%**
Rubella	242	0.04	99.98%	0.09	**100%**
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

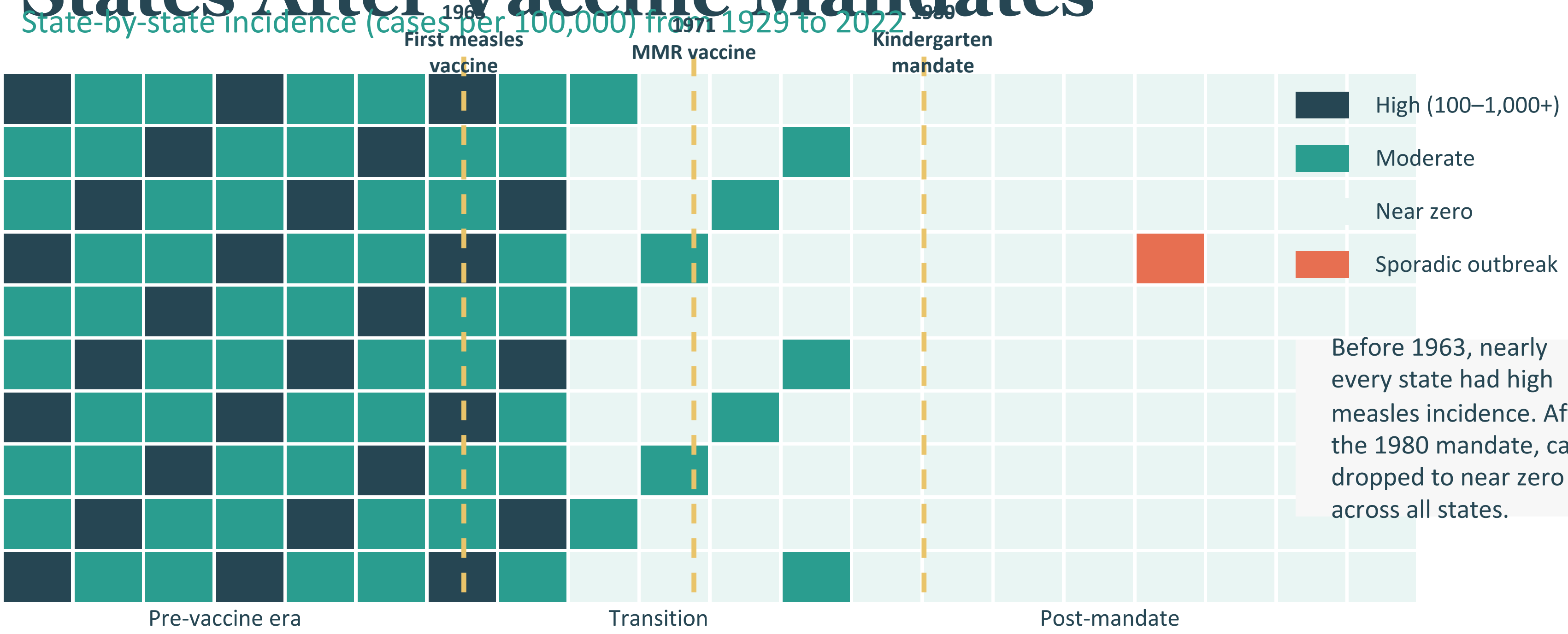
= 100% elimination

= Incomplete reduction

Source: Roush and Murphy (2007), via Our World in Data

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

State-by-state incidence (cases per 100,000) from 1929 to 2022



Before 1963, nearly every state had high measles incidence. After the 1980 mandate, cases dropped to near zero across all states.

Source: Our World in Data

Smallpox: The Only Human Disease Eradicated Through Vaccination

1796
First vaccine (Jenner)

1959
WHO resolution to eradicate

1967
Intensified campaign

1977
Last natural case (Somalia)

1980
Certified eradicated

Why smallpox was uniquely suited for eradication:

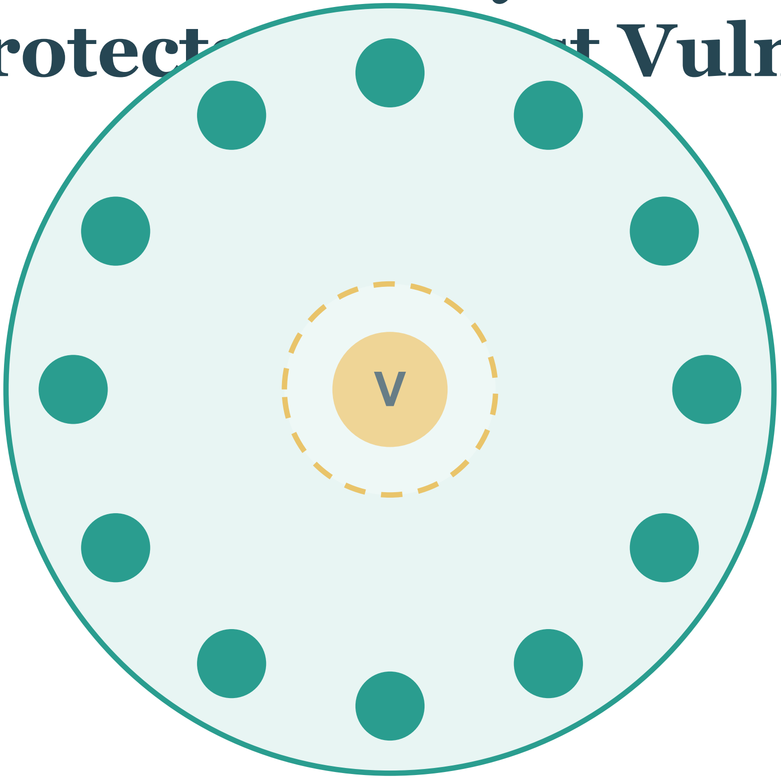
- Infected only humans — no animal reservoir
- Symptoms were clearly distinguishable
- Vaccine was cheap, stable, and highly effective

1980

Smallpox declared eradicated worldwide — the only human disease fully eliminated through vaccination

Source: Our World in Data

Herd Immunity: How Vaccinating Many Protect t Vulnerable



Vaccinated individuals

Vulnerable individual

Source: Our World in Data

How it works

When enough people are vaccinated, pathogens cannot spread effectively through a population.

This creates a protective barrier around those who cannot be vaccinated:

- Newborns too young for vaccines
- Immunocompromised individuals
- Cancer patients on chemotherapy

Vaccines protect not only individuals but entire communities.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal

40.5%
case reduction

31.3%
death reduction

Hepatitis A

89%
case reduction

88.7%
death reduction

Acute Hepatitis B

83.9%
case reduction

83.6%
death reduction

Barriers to full coverage

Access gaps

Not everyone has access to vaccines globally

Supply shortages

Shortages put children at risk in low-income regions

Trust challenges

Poor scientific understanding reduces uptake

Source: Our World in Data

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

1. Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US
2. Vaccine development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA and genomic tools
3. Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate
4. Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign
5. Coverage gaps persist for pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%)

Call to action: Improve global access, coverage, and trust in vaccination to close remaining gaps and protect vulnerable communities.

Source: Our World in Data